Tutorial 8 PL / SQL contd. (Guide)

Tutorial Objectives:

- **Continue to learn PL / SQL**
  - **Processing with Cursors**
  - **Write Triggers**

Tutorial Tasks:

Cursors:

1 Provide a list of customers with various customer balances.
Use cursor attributes:
%FOUND or %NOTFOUND
%ROWCOUNT
%ISOPEN

a. List customers who have a balance of zero
b. List customers who have a balance of over 500

You can either write a PL/SQL block or a stored procedure but must use a cursor and cursor attributes. Book discusses these under 8-7c make sure you understand why these are needed.

Based on the customer and invoice tables (Tutorial 7):

a. No records are returned.
b. 3 customers meet the condition.

2 Provide a list of customers who have invoices.
Use cursor attributes:
%FOUND or %NOTFOUND
%ROWCOUNT
%ISOPEN

You can display multiple records for a customer along with their invoices (e.g., invoice number, invoice amount, etc.).
For example, Customer 1000 – has 3 invoices.

3 Provide a list of customer names as follows:
If a customer has no invoice, then the customer's name should follow with "-- no invoice"
If a customer has invoices, then the customer's name should follow with "-- has invoice"
Use cursor attributes:
%FOUND or %NOTFOUND
%ROWCOUNT
%ISOPEN

Consider finding (i.e., counting) whether a customer has zero or more invoices to capture the two given situations.
Check your code works for 'No invoice' option. For example, modify your Invoice Table data by deleting INV_NUM = 8008, then your database will have a customer 1002 with no invoice. Alternatively, add a new customer but no corresponding invoice.

4 Rewrite at least one of the above cursors using a FOR loop (if not already used).

Compare your earlier code and notice how with FOR loop your code length drops significantly.

Triggers:

Section 8-7a (pg. 401) of your textbook and week8 lecture talks about triggers. Make sure you understand the syntax and the two trigger types - statement and row-level.

5 Write a trigger to update the customer balance in the CUSTOMER table when a new invoice record is entered. Assume that the sale is a credit sale.

Name the trigger **TRG_UPDATE_CUST_BAL_INS**.

Test the trigger using the following new INVOICE record:
8089, 1001, 17-Apr-2022, 225.40

Assuming you have not changed the database (especially for customer 1001), the customer balance for customer number 1001 should change from 840.92 to 1066.32.

6 Write a trigger to update the customer balance in the CUSTOMER table when an invoice is deleted.

Name the trigger **TRG_UPDATE_CUST_BAL_DEL.**

Test your trigger using the procedure **PRC_INV_DEL** (written earlier in Tutorial 7).

7 Write a trigger to record the user and date when a new customer record is added, updated, or deleted.

Note: You can store this information in another table (e.g., CUST_AUDIT). You will first need to create this table with the required columns.

You can use auto-generated sequence to add the Primary Key values in CUST_AUDIT table.